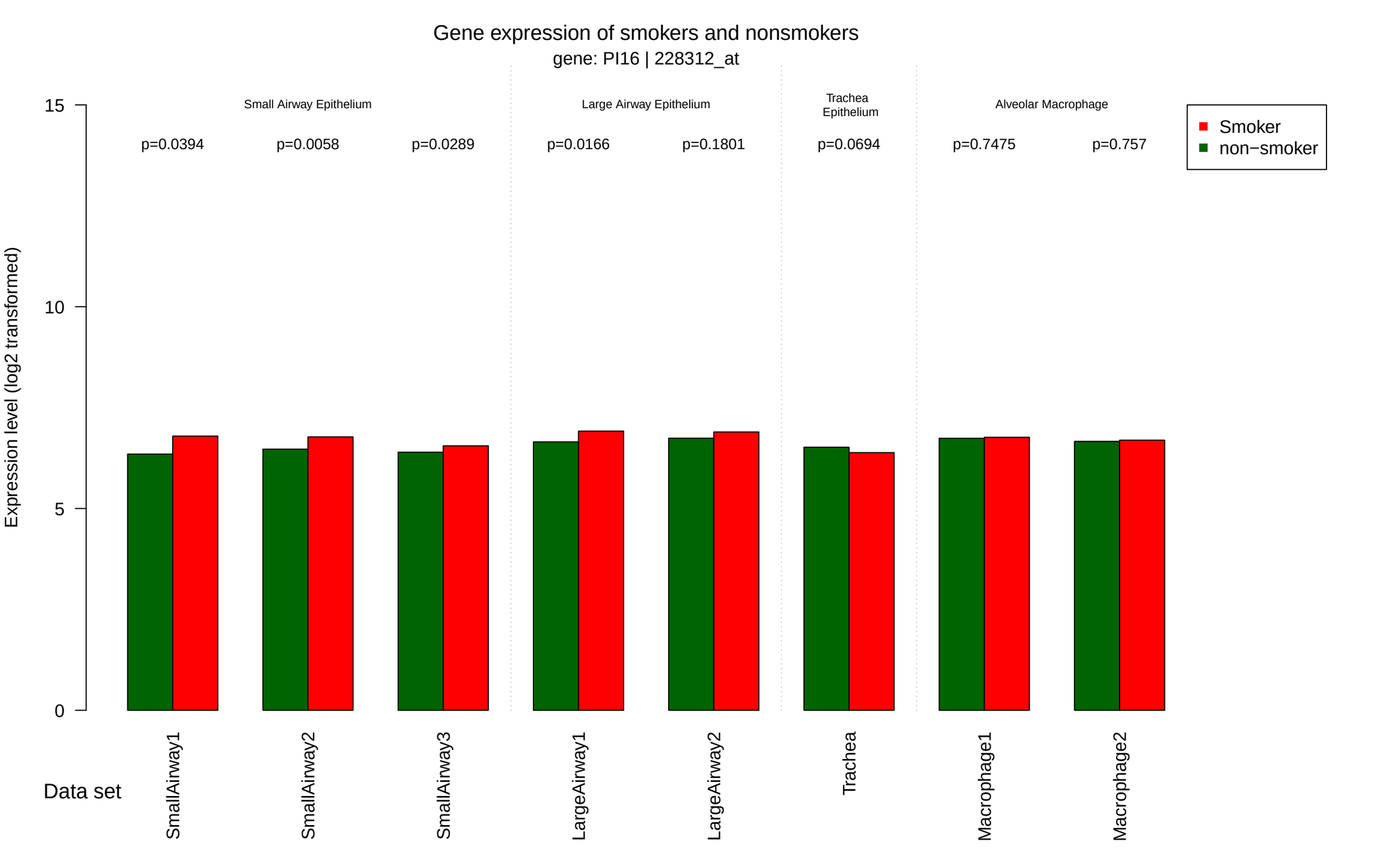
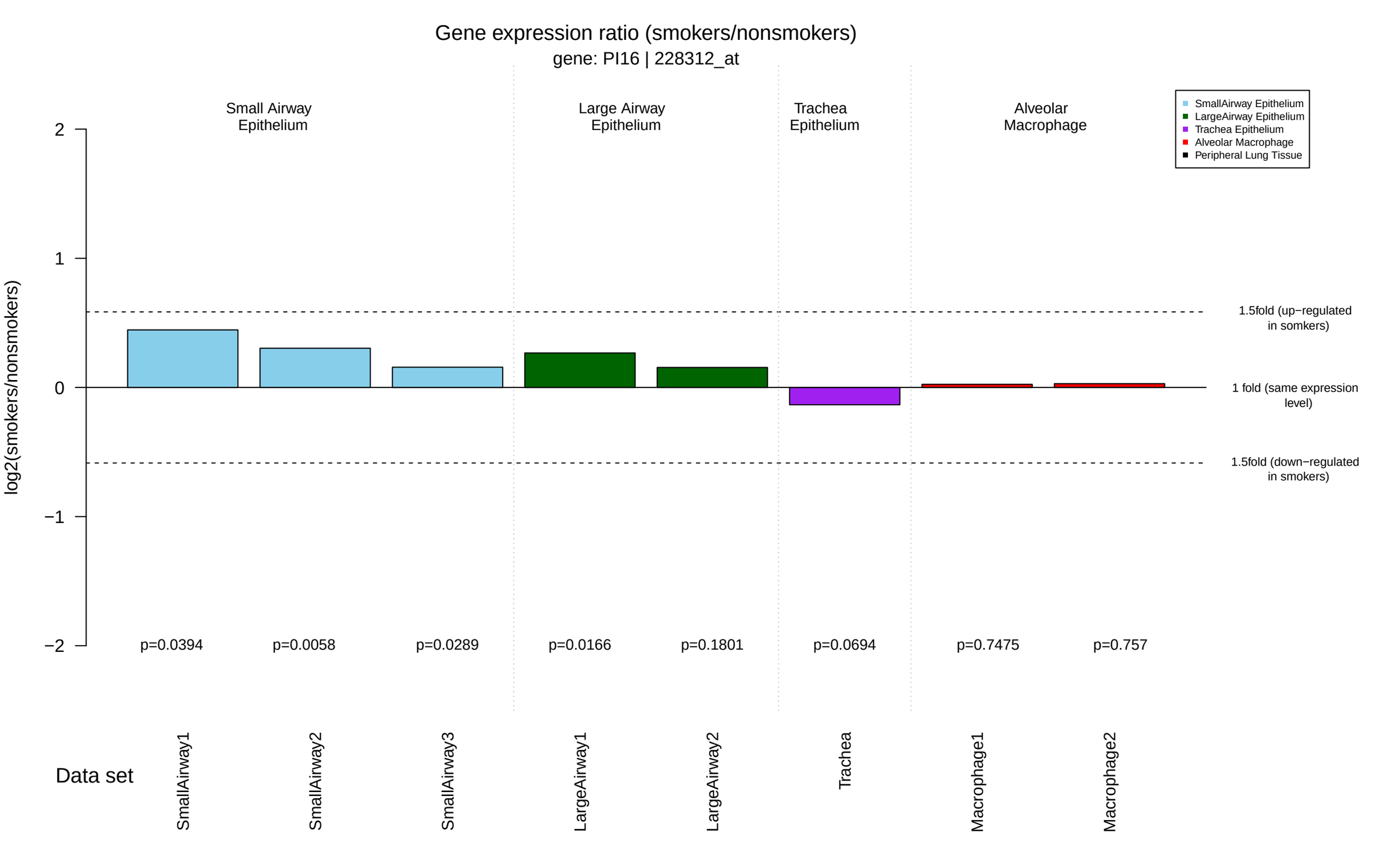
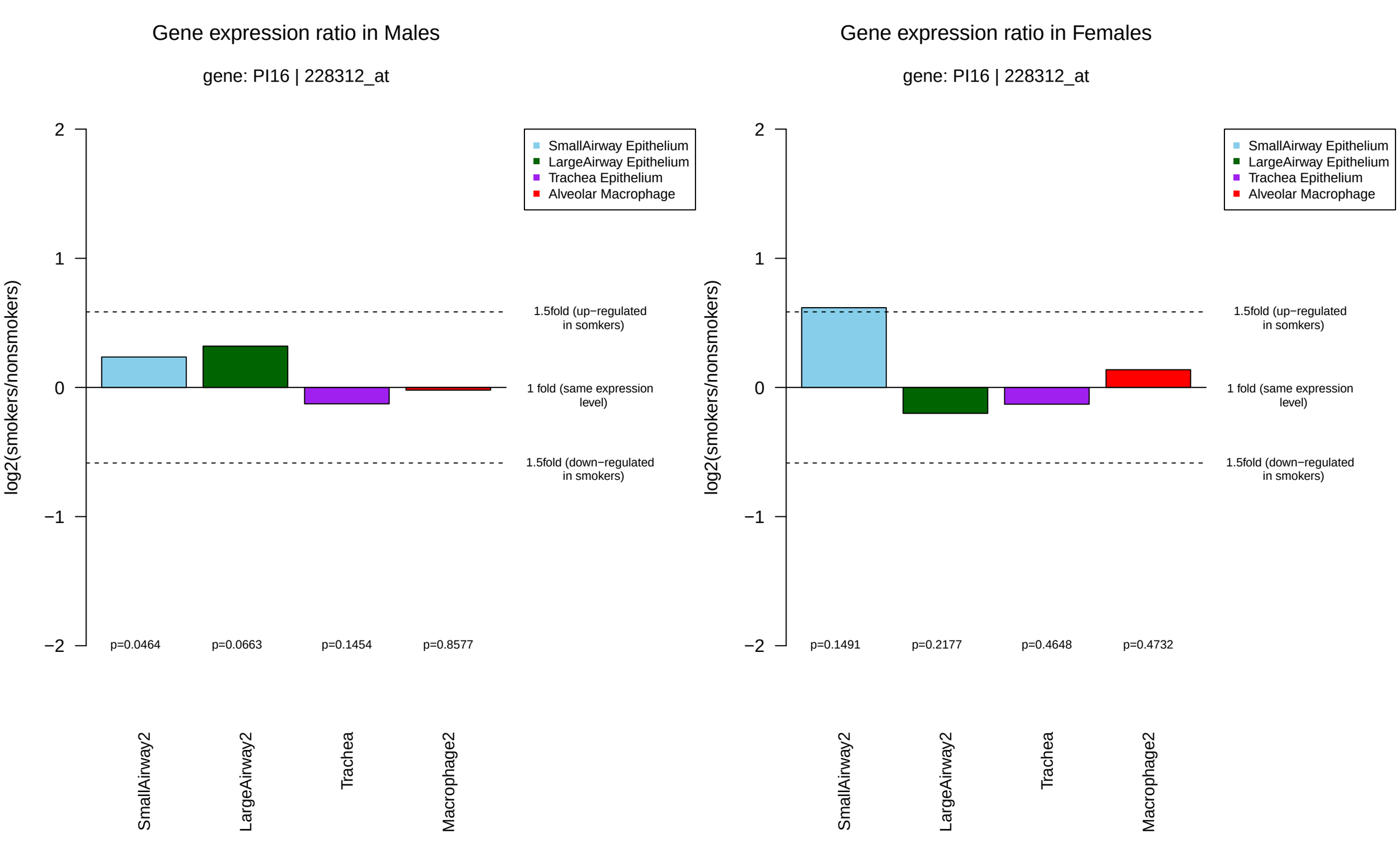
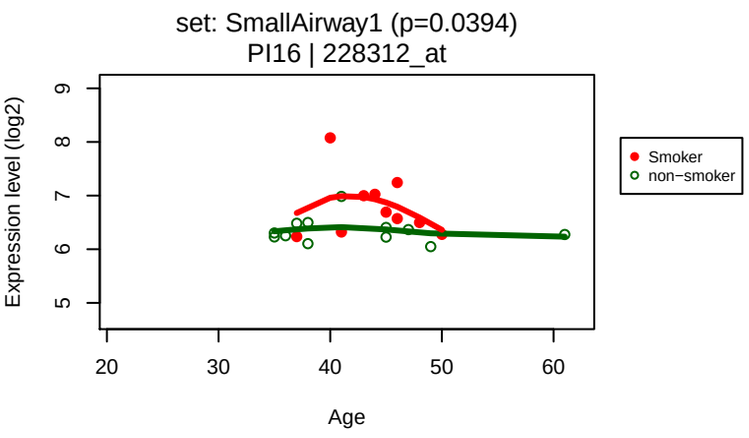
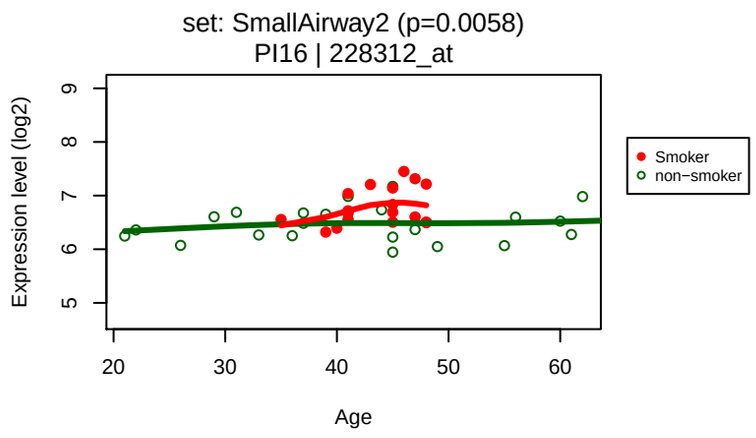
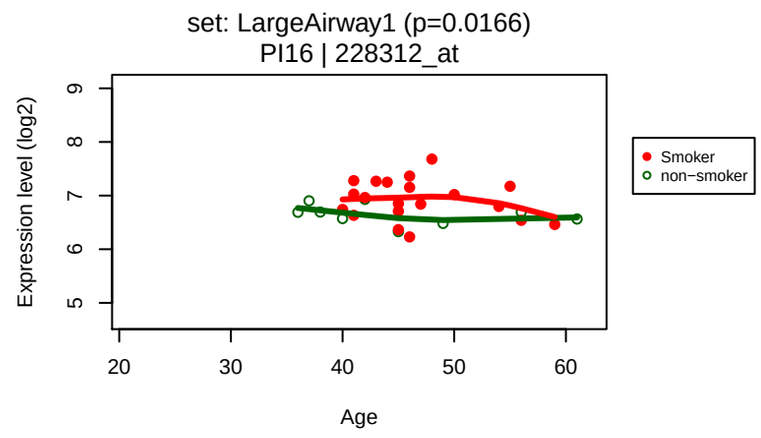

Gene expression of smokers and nonsmokers
gene: PI16 | 228312_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0394
p=0.0058
p=0.0289
p=0.0166
p=0.1801
p=0.0694
p=0.7475
p=0.757
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

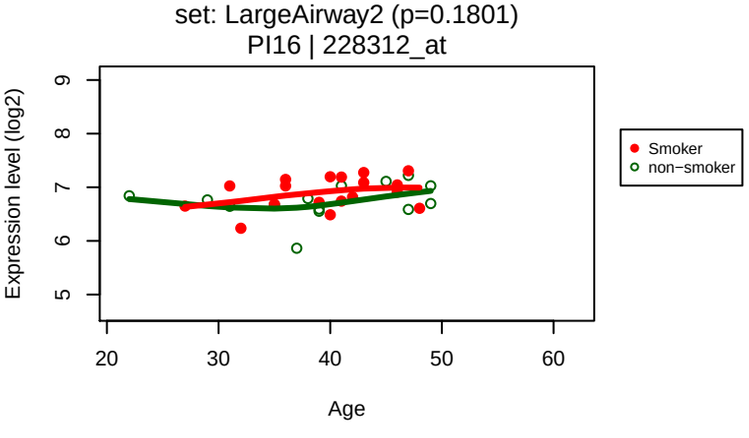

Gene expression ratio (smokers/nonsmokers)
gene: PI16 | 228312_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0394
p=0.0058
p=0.0289
p=0.0166
p=0.1801
p=0.0694
p=0.7475
p=0.757
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

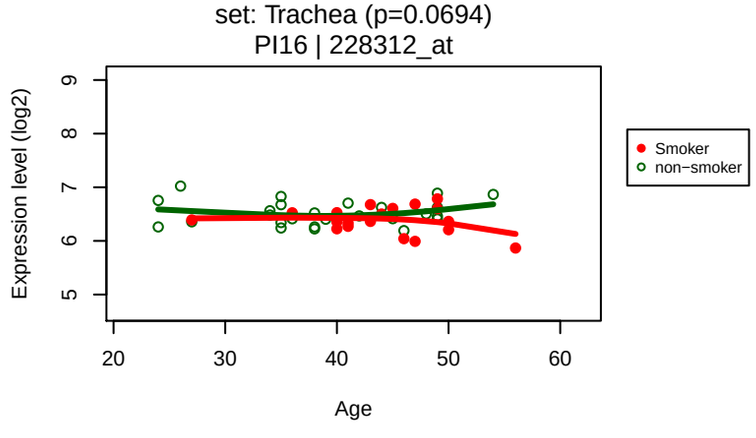

Gene expression ratio in Males
gene: PI16 | 228312_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0464
p=0.0663
p=0.1454
p=0.8577
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PI16 | 228312_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1491
p=0.2177
p=0.4648
p=0.4732
SmallAirway2
LargeAirway2
Trachea
Macrophage2

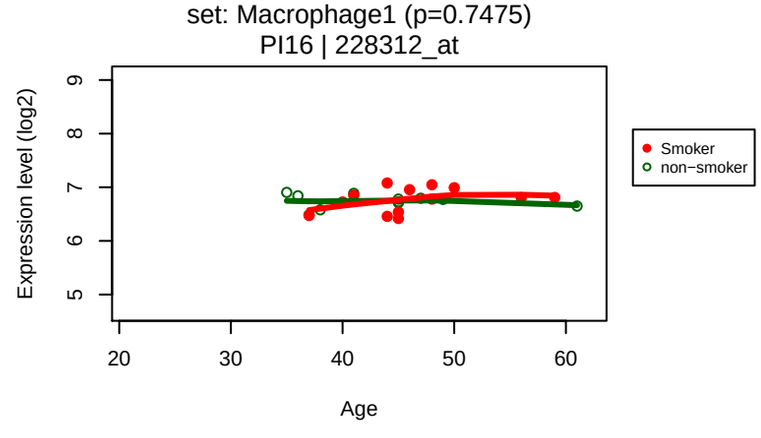

set: SmallAirway1 (p=0.0394)
PI16 | 228312_at
Expression level (log2)
Age
Smoker
non−smoker

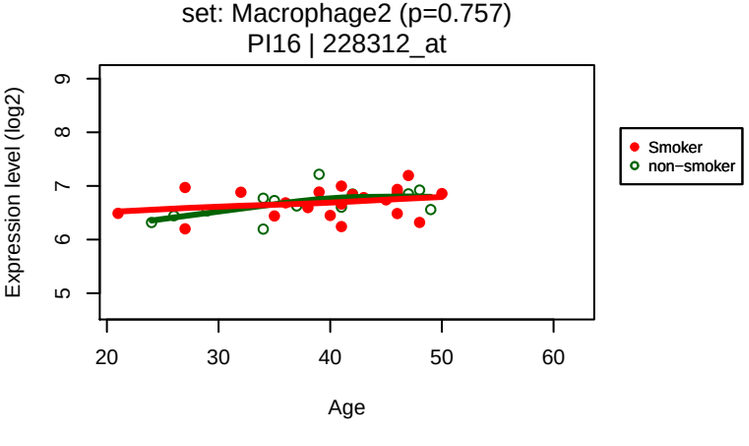

set: SmallAirway2 (p=0.0058)
PI16 | 228312_at
Expression level (log2)
Age
Smoker
non−smoker

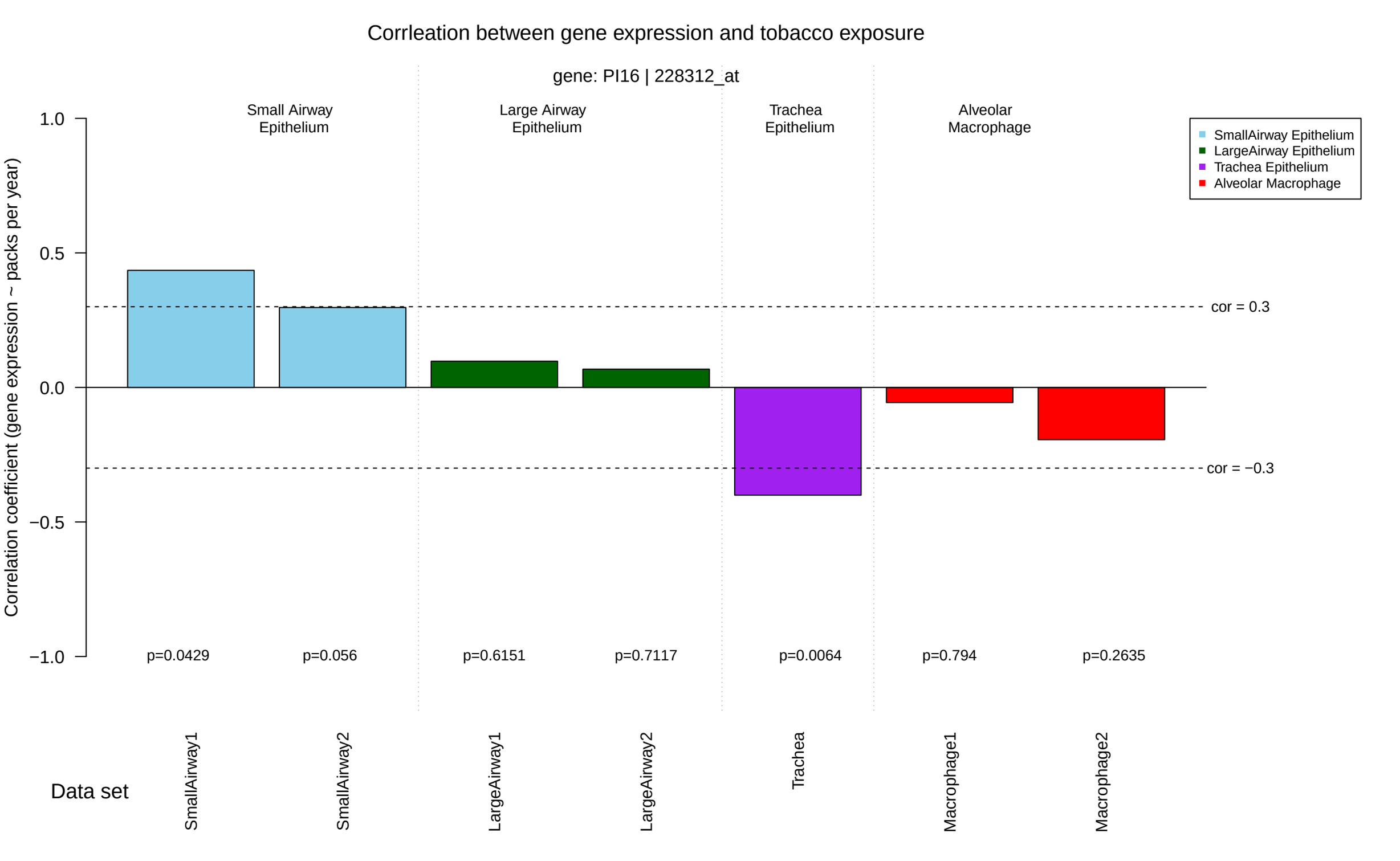

set: LargeAirway1 (p=0.0166)
PI16 | 228312_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1801)
PI16 | 228312_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0694)
PI16 | 228312_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7475)
PI16 | 228312_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.757)
PI16 | 228312_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PI16 | 228312_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0429
p=0.056
p=0.6151
p=0.7117
p=0.0064
p=0.794
p=0.2635
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2